AR floor speaker preview

Tair Asset · AIB 2401 · Student ID 242458
AR VR XR Applications · Practical assignment 3

This page places a 25 × 30 × 100 cm speaker on a real surface. ID 242458 selects single placement, a 12 degree tilt limit and a front that faces the user at placement. A second valid tap replaces the first placement.

Device evidence pending. Geometry and simulated logic have been checked. No compatible-device recording, physical measurements or device screenshots are available yet. This document is a technical draft, not a completed field report.

E1 Reference spaces

Prediction. Walking about one metre should change the camera position in local space by roughly that displacement. A fixed hit should have different viewer-relative coordinates as the phone moves.

Result. The camera positions and paired hit poses have not been recorded on a compatible device. Capture pose exports all three from the same active frame.

Explanation. Viewer space follows the phone, so it gives the hit-test ray its direction. Local space provides the scene coordinate system for both the hit and the rendered object. A pose describes the same point relative to the requested reference space; different coordinates do not imply different physical hits. [1]

E2 Hit results and persistent attachment

Prediction. After a three-metre walk away and back, anchor-based attachment may follow map corrections better than a frozen local matrix. Neither method guarantees zero drift.

Result. Return drift is unmeasured. Simulated checks verify anchor deletion and asynchronous cleanup, not real tracking stability.

Explanation. A hit result is a frame-specific intersection, not an automatically saved spatial anchor. PA3 requires a separate createAnchor call and per-frame anchor pose updates. Slide 16 also distinguishes transform placement from anchors. Local-floor changes the origin to estimated floor height; changing coordinates does not improve hit testing. [1, 2]

The PDF names three tasks: GL/layer compatibility, setReferenceSpaceType, and setSession (Ch. 7, pp. 9–11); slide 9 groups them differently. In three@0.186.1, setSession handles task 1: conditional makeXRCompatible (427–431) and XR layer/render state (451–453 or 496–498). Task 2 remains explicit: setReferenceSpaceType('local'); its setter is 311–315, and setSession requests that chosen type at 525–526. Task 3 is our awaited setSession call, which also starts XR animation (528–529). Therefore no separate setupWebGLLayer is needed. [1–3]

E3 Reading the surface pose

Prediction. A normal within 12 degrees of world up permits placement; a steeper normal produces a red ring and rejects the tap.

The pose matrix is stored column by column

Xx	nx	Zx	px
Xy	ny	Zy	py
Xz	nz	Zz	pz
0	0	0	1

Normal n = elements 4, 5, 6

Position p = elements 12, 13, 14

tilt = acos(ny / length(n))

Accept when tilt <= 12 degrees

Result. The two required device screenshots and one measured matrix are pending. Automated rotations at 0, 8, 12, 16, 45, 90 and 180 degrees passed the mathematical filter check.

Explanation. The Y column is the surface normal. Its dot product with world up equals its Y component. Normalise, clamp the cosine and apply acos. The exact 12 degree boundary is accepted; 12.001 degrees is rejected. The position column does not determine tilt.

E4 Surface detection

Prediction. A textured surface should usually offer more stable visual features than a plain white or reflective surface. This is a hypothesis, not a measured ordering.

Result. No compatible-device surface trials have been completed.

Surface	First hit seconds	Success
Patterned	Not measured	Not tested
Plain white	Not measured	Not tested
Glass or glossy	Not measured	Not tested

Explanation. E4 asks us to connect camera feature points, motion sensors and visual-inertial odometry: visual references and motion measurements jointly estimate device movement, supporting surface detection. Our code receives the resulting poses and intersections, not the feature detector itself. Surface-dependent differences must be explained from observations; low texture and reflections are hypotheses to test. Our timing protocol starts at source readiness and records no hit within 20 seconds as failure. [2]

E5 Scale check

Prediction. A one-metre geometry should appear one metre tall, within the uncertainty of tracking and measurement.

Result. Box3 measures 0.25 × 1.00 × 0.300000005 m and a base at y=0. Physical apparent height and height error remain unmeasured. The code dimensions cannot substitute for the required tape/reference-object test.

Explanation. A biased surface or tracking estimate can change apparent placement. A reference ruler at a different depth, perspective or an imprecise reading can also distort the comparison. The reference should stand next to the product at the same depth.

E6 Retail implications and privacy

Device-specific failure observations remain pending. Two risks to test are a delayed reticle, which could make a shopper abandon the preview, and return drift, which could undermine confidence in fit. A scan-progress prompt and a rescan option are candidate fixes. For a Kazakhstani retailer, fit claims should include measured uncertainty. This implementation reads device poses, hit transforms and anchor poses. It neither reads raw camera pixels nor requests a full room mesh or feature-point map. Seeing the camera background does not mean our script reads its pixels. Tell users why tracking permission is needed and that exported diagnostics contain spatial coordinates. The page does not upload camera images or pose logs. This describes our requested features, not every possible WebXR extension.

Development log

The first preview hid the light driver membranes behind the surround caps.
Visual inspection and component depth comparison identified the overlap.
The membrane depth was corrected and checked against bounds in commit e10f7e2.
Simulated frame tests also cover replacement, Clear and delayed anchors.

Sources

1. Supplied Baruah Ch. 7 PDF: pp. 7–11, 14–19; supplied Ch. 7 slides: 8–9, 11–16.
2. Supplied PA3 assignment: Tasks A–E and write-up E1–E6.
3. Pinned three@0.186.1 WebXRManager.js, inspected as E2(c) requires: versioned source.
4. Supplied Ch. 5 PDF: pp. 7–10, 26, 35; Ch. 6 PDF: pp. 9–10, 24–27.